

2023 AMC 10A Problem 3 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 3

Problem:

How many positive perfect squares less than 2023 are divisible by 5?

Answer Choices:

- A. 8
- B. 9
- C. 10
- D. 11
- E. 12

Solution:

Because 5 is prime, a perfect square is divisible by 5 if and only if its square root is divisible by 5. Note that $40^2 = 1600 < 2023 < 2025 = 45^2$, so $5^2, 10^2, 15^2, 20^2, 25^2, 30^2, 35^2$, and 40^2 are the only positive perfect squares satisfying the given condition. There are (A) 8 such numbers.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) [↗](#)

